Deep Learning Model for the Prognostic Prediction of Acute Inflammation of the Central Nervous System Using T2-weighted Brain Magnetic Resonance Imaging

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Background

- Acute inflammation of the central nervous system (CNS) caused by various pathogens is associated with high fatality rates and disability.
- The causes of acute CNS inflammation are diverse, ranging from infectious agents, making early identification of the underlying cause and prognosis and prompt treatment particularly challenging.
- We used a deep learning model for the prognostic prediction of CNS acute inflammation using brain MRI data.

Methods

- T2-weighted brain MRI images (T2WI) were collected retrospectively from patients with acute CNS inflammation who were admitted to a tertiary referral hospital between January 2010 and December 2023.
- Images obtained from 2010 to 2019 were used as the internal training dataset, while those from 2020 to 2023 served as the external dataset.

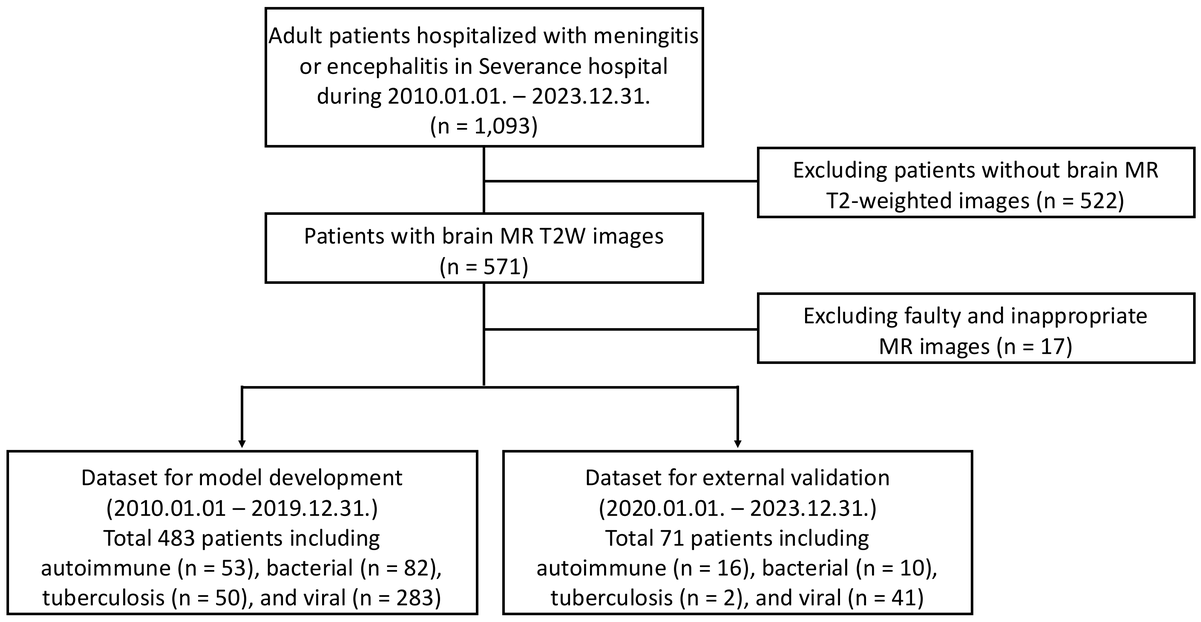

Figure 1. Patient selection flowchart

- The preprocessing was performed using the Python FastSurfer library. Through this process, the 2D T2WI was converted into a 3D NumPy array with dimensions of 560 × 560 × 24.
- Prognosis prediction was performed by categorizing patients based on their modified Rankin Scale (mRS) scores at discharge, with scores of 3 or higher classified as poor outcomes and scores below 3 as good outcomes.
- The MRI data of the entire patient cohort were used for training, and to account for differences in prognosis based on the cause of inflammation, the data were divided into four groups, with training conducted separately for each group.
- For deep learning training, a modified 3D DenseNet architecture was utilized and the performance of the model was assessed using accuracy and area under the receiver operating characteristic curve (AUROC).
- All processes were conducted on a computing workstation equipped with an Intel Xeon Gold 6326 CPU (2.90GHz, 16 cores) and two NVIDIA A40 GPUs with 46GB of memory each. The preprocessing of brain MRI and the development of deep learning models were carried out using Python version 3.10.4 and PyTorch version 1.13.1.

Results

- A total of 570 patients (483 for training and 70 for validation) from Severance Hospital in Repulic of Korea were included in the study.
- When categorized by pathogen, viruses were the most common, followed by bacteria and autoimmune causes (Figure 1).
- The differences in initial clinical information between the good and poor prognosis group are detailed in Table 1.

Results

Table 1. Comparison of initial clinical information of CNS acute inflammation between each prognostic group

Characteristics	Good (n = 480)	Poor (n = 91)	P-value
Female, n (%)	221 (46.0%)	42 (46.2%)	1.000
Age, mean ± standard deviation	46.9 ± 19.0	56.1 ± 19.8	< 0.001
Abnormal mentality, n (%)	140 (29.2%)	60 (65.9%)	< 0.001
Vital signs			
Mean systolic BP (mmHg)	129.4 ± 16.2	133.5 ± 18.7	0.050
Mean body temperature (°C)	37.3 ± 0.6	37.3 ± 0.5	0.969
Cerebrospinal fluid findings			
White blood cell counts (/μL)	437.3 ± 1478.9	229.2 ± 657.8	0.005
Protein (mg/dL)	132.6 ± 149.0	313.6 ± 819.6	0.294
CSF/serum glucose ratio	0.5 ± 0.2	0.5 ± 0.2	0.656

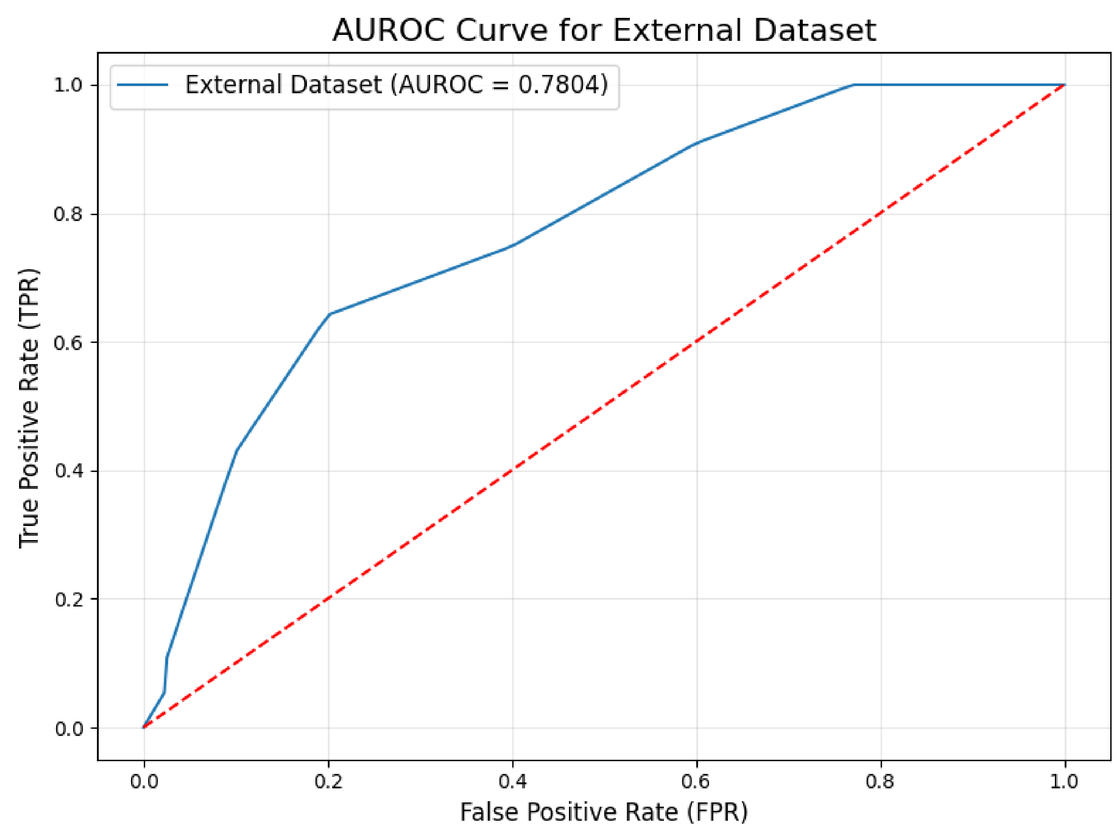

Figure 2. Receiver operating characteristic curve of prognosis prediction model

- After training on the entire patient cohort using the deep learning model, testing on the external dataset yielded an AUROC of 0.7804 and an accuracy of 0.8231 (Figure 2).
- When patients were classified by pathogen and separate models were built and trained for each group, the results showed AUROC values of 0.8737 for autoimmune, 0.6667 for bacterial, 1.0000 for tuberculosis, and 0.6513 for viral cases (Table 2).

Table 2. The prognostic performance of deep learning model of each etiologic group

	Autoimmune	Bacteria	Tuberculosis	Virus
AUROC	0.8737	0.6667	1.0000	0.6513
Accuracy	0.8333	0.5417	1.0000	0.7444
Poor outcome	4 (25.0%)	1 (10.0%)	1 (50%)	2 (4.9%)

Conclusion

- Our deep learning model, which uses T2-weighted 3D brain MRI data, could predict the prognosis of acute inflammation of the CNS.
- It is anticipated that integrating clinical information to develop a multimodal deep learning model will enable more accurate and practical predictions.

Acknowledgement

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